Handin Random Forest analyses air quality

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Task

Concentrations of air quality components are affected by meteorology. A random forest model can be used to estimate the effect of the meteorology on the concentrations. Build a random forest model for an air quality component based on meteorological parameters (wind, temperature, etc.), train the model with observations from 2016 to 2018 and make a prediction with the model for 2019. Test the performance of the model for 2019 by comparing model predictions with the observations for 2019.

Steps

1. Select one air quality concentration (NO_2, NOx, O_3, PM_{10}, or $PM_{2.5}$) at one station (street: NL10236, NL10636 or NL10639, urban: NL10247, NL10418 or NL01488, or rural: NL10617, NL10644 or NL10738). You may also select another component, but check if there is sufficient data available.
2. Select the following relevant explanatory variables (wd, ws, t, q, hourly_rain, p, n , rh) (see Table 1). You can also define parameters yourself such as, hour of the day, day of the week, day of the year, etc.
3. Read the data from file in python.
4. Build a random forest model using all 8 meteorological variables (see an example of the structure of code in python below). n_estimators = number of trees (use 30 initially)
5. Train the model with data from 2016 to 2018. Think about what to do with missing data and negative concentrations and discuss your choice in the extended abstract.
6. Make a prediction with the model for 2019.
7. Plot the observed and predicted concentrations. Do you understand what you see?
8. Calculate statistics for the agreement between the observed and predicted values (correlation coefficient, rmse, etc.). Do this for the training period (2016-2018) and 2019.
9. Investigate the effect of varying number of trees and/or other parameters on the prediction.
10. Investigate which meteorological variable explain most of the data. Do this, e.g., by 8 runs with each time a different (single) meteorological variables. Investigate the effects in terms of meteorology and chemistry of the atmosphere.
11. Optional: Check how the model performs for another location.

For the hand-in

12. The written hand-in should be in the form a extended abstract with at the top a (meaningful) title and your name (and student number). Describe in the body of the text: what is being studied and why, a short description of the methods and data, a discussion of the results and conclusion. Also add up to 3 references.
13. The extended abstract should be **no more than 2 A4 pages and not more than 900 words** (including everything). **Include not more than 2 figures** (select the ones that tell the main story).
14. Also submit the computer code as an appendix (code will not be graded).
15. If an AI tool, such as CHATGPT is used for coding, than this must be acknowledged in the abstract. You are yourself responsible for what you hand in, so make sure the code is correct.

Be concise. All elements need to be in the abstract, but the focus should be on the results and discussion in term of physics and chemistry.

➔ **The use of AI tools, such as ChatGPT is not allowed for the extended abstract**.
Remember: Use of ChapGPT is fraud.

For the grading I will pay special attention to the logic of arguments and a good description of your results.

The grading will be based on:

- Completeness of the abstract, all elements discussed (25%)
- Discussion of the results (50%)
- Layout of the paper (figures, legends, units, etc.) (25%)

After the abstract is handed in, I will grade them. After that, I will talk with each of you individually about the abstract. The goal is to see that you can explain and understand what you wrote in the abstract. If you can than the final grade is the same as the grade of the abstract. If this is not the case up to 3 points are subtracted from the grade. If you do a very good job in explaining everything 1 point can be added to the grade.

With a total grade lower than 5.0, a retake is necessary. Maximum grade of the retake if 6.0

Data files (street, urban and rural location)

- Street_NL10636_year.csv, Street_NL10639_year.csv (Utrecht), and Street_NL10236_year.csv (Eindhoven)
- Urban_NL10418_year.csv, Urban_NL01488_year.csv (Rotterdam), and Urban_NL10247_year.csv (Veldhoven)
- Rural_NL10644_year.csv (Cabauw), Rural_NL10738_year.csv (Wekerom), and Rural_NL10617_year.csv (Biddinghuizen)

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#==========================================
# First make sure sklearn is imported (needed to be done only once)
import sklearn

#==========================================
# Random Forest code structure

from sklearn.ensemble import RandomForestRegressor

# Read in the data for training (x_train, y_train for 2016-2018) and for prediction (x_pred for
2019)
#x_train(n_data, n_variables)  = 2d-array with explanatory variables
#y_train(n_data)                = 1d-array with observations
#x_pred(n_data2, n_variables)= 2d-array with explanatory variables

# Define the Random Forest model
model = RandomForestRegressor(n_estimators=30, min_samples_leaf=5, random_state=42)

# Fit the Random Forest model on the whole dataset
model.fit(x_train, y_train)

# Determine the R2 score of the model
score = model.score(x_train, y_train)
print('R2 score: {:6.3f}' .format(score))

# Make a prediction for x_pred for 2019
y_pred = model.predict(x_pred)

# Plot data and Calculate statistics by comparing y_pred with observation for 2019
#==========================================
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Table 1. Explanation variables in data file

date	Date/time in UCT+1 (NL winter time)
wd	Wind direction (in degrees) average over the last 10 minutes of the past hour (360=north, 90=east, 180=south, 270=west, 0=no wind 990=variable
ws	Wind speed hourly average (in m/s)
ff	Wind speed (in m/s) average over the last 10 minutes of the past hour
fx	Highest wind gust (in m/s) of the past hour
t	Temperature (in Celsius) at 1.50 m observation height
t10n	Minimum temperature (in Celsius) at 10 cm in the past 6 hours
td	Dew point temperature (in Celsius) at 1.50 m observation height
sq	Sun shine (in hours) of past hour, calculated from global radiation (-1 for <0.05 hour)
q	Global radiation (in J/cm2) per hour
dr	Precipitation (in hours) per hour
hourly_rain	Precipitation (in mm) (-1 for <0.05 mm)
p	Pressure (in hPa) at sea level
vv	Horizontal sight (0=less than 100m, 1=100-200m, 2=200-300m,..., 49=4900-5000m, 50=5-6km, 56=6-7km, 57=7-8km, ..., 79=29-30km, 80=30-35km, 81=35-40km,..., 89=more than 70km)
n	Cloud cover (in 1/8), (9=air above invisible)
rh	Relative humidity (in %) at 1.50 m observation height
bc	Black carbon (µg m^{-3})
co	Carbon monoxide (µg m^{-3})
nh3	Ammonia (µg m^{-3})
no	Nitrogen monoxide (µg m^{-3})
no2	Nitrogen dioxide (µg m^{-3})
nox	Nitrogen oxides (in µg NO_2 m^{-3})
ox	Odd oxygen (µg m^{-3})
o3	Ozone (µg m^{-3})
pm10	Particulate matter < 10 µm (µg m^{-3})
pm25	Particulate matter < 2.5 µm (µg m^{-3})
so2	Sulfur dioxide (µg m^{-3})
benzeen	benzene (µg m^{-3})
h2s	Hydrogen disulfide (µg m^{-3})
ufp	Ultra fine particles (µg m^{-3})